

Module2Lab2

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1 Module 2 Lab 2: Textbook Problem 4.2

This notebook demonstrates how multiplication can be performed through repeated addition rather than the multiplication operator. It includes two implementations: one using a for loop and another using a while loop.

```
[1]: def multiply_with_for(number, repetitions):  
    result = 0  
  
    for count in range(repetitions):  
        result = result + number  
  
    return result
```

```
[2]: def multiply_with_while(number, repetitions):  
    result = 0  
    count = 0  
  
    while count < repetitions:  
        result = result + number  
        count = count + 1  
  
    return result
```

```
[3]: first_number = 7  
second_number = 6  
  
for_result = multiply_with_for(first_number, second_number)  
while_result = multiply_with_while(first_number, second_number)  
real_product = first_number * second_number  
  
print("For-loop result:", for_result)  
print("While-loop result:", while_result)  
print("Python product:", real_product)  
print("Both functions agree:", for_result == while_result == real_product)
```

For-loop result: 42

While-loop result: 42

```
Python product: 42
Both functions agree: True
```

1.1 Conclusion

Both functions correctly returned 42 for $\{7 \times 6\}$. The for-loop and while-loop implementations achieve multiplication by repeatedly adding 7 a total of six times. This illustrates how loops can accumulate a running total and ultimately return the final result.